Stick-together calculations

antocciphysics | TEACHER KEY | CPA

Learning target: Find a common final velocity using total momentum.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 3, Day 5; 3 course days later.*
15 kg·m/s right, so the vector sum remains zero.

02. Explain a worked example (worked)
Both bodies share one final velocity; total mass is 3 kg and total momentum remains 6.

03. Look and explain (model)
Both masses because both share final velocity.

04. Build the solution (practice)
6 kg·m/s; 3 kg; 2 m/s right.

05. Use the idea (reason)
When the masses are equal; generally use a mass-weighted average.

06. Choose a method (mixed) *Review: Unit 2, Day 12; 12 course days later.*
No. Tangent velocity arrows change direction, so inward acceleration and net force are needed.

07. Exit check (exit)
0 m/s.

08. STRETCH: Apply it another way (stretch)
Total mass 12/2=6 kg, so stationary cart mass 4 kg.

Stick-together calculations

antocciphysics | TEACHER KEY | HONORS

Learning target: Find a common final velocity using total momentum.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 3, Day 5; 3 course days later.*
Other velocity -6 m/s; total K=240 J, supplied by released internal energy.

02. Explain a worked example (worked)
Both bodies share one final velocity; total mass is 3 kg and total momentum remains 6.

03. Represent and interpret (model)
Both masses because both share final velocity.

04. Solve and justify (practice)
p=7; vf 1.4 m/s; initial KE 26.5 J; final 4.9 J; transformed 21.6 J.

05. Analyze the reasoning (reason)
When the masses are equal; generally use a mass-weighted average.

06. Choose a method (mixed) *Review: Unit 2, Day 12; 12 course days later.*
Both become 1/4; $a_c=v^2/r$ and $F_{net}=mv^2/r$.

07. Exit check (exit)
3 m/s; KE 9 J versus 90 J, 10% remains.

08. STRETCH: Apply it another way (stretch)
First total momentum zero, final v=0. Second total 4 kg·m/s, final v=4/3 m/s right.